Sea Winds Predictions, Phase 2 Report

Michael Ibrahim

Executive summary

This report presents one integrated solution to the competition's forecasting, wind-farm siting, and forecast-value tasks. The forecast estimates wind-speed and wind-direction uncertainty on the 1.3 km target grid at one, seven, and fourteen days. The siting procedure selects a legal centre and a 55-turbine layout for the fixed 1.21 GW IEA 22 MW farm. The same wind forecast is then converted into six-hour farm-energy quantiles for decision making under asymmetric imbalance costs.

The forecasting method begins with the supplied high-resolution atmospheric forecasts. Wind speed is calibrated with lead-specific gradient-boosted quantile regressors, while direction is corrected as a circular vector residual so that values near 0°and 360°remain close. A two-model terrain downscaler predicts the eastward and northward wind components at target resolution from interpolated coarse wind, location, elevation, and distance to shore. Additional predictors are limited to lagged information available at the forecast issue time and trajectories from GraphCast, a global weather model trained on ERA5. They are used only for leads, dates, hours, and physical conditions in which paired historical tests show a stable improvement. The resulting artifact contains 29 fitted estimators and is generated with one single-threaded training process.

The definitive forecast contains all 4,196,640 required rows. The wind-speed interval scores are 9.511, 17.124, and 16.835 at one, seven, and fourteen days. The corresponding wind-direction scores are 76.400, 335.571, and 315.183, and the rank-based primary score is 1.497276. These results show strong short-range directional skill and useful long-range speed calibration, while long-range direction remains the main forecast limitation.

The selected farm centre is 54.109676°N, 0.952316°E at a supplied-grid depth of 44.14 m. The submitted layout satisfies every geographic, depth, footprint, turbine-count, and spacing requirement. Five-year PyWake replay gives a mean capacity factor of 50.36%, mean annual energy production of 5,337.7 GWh, and mean wake loss of 7.10%. Relative to a valid rectangular layout at the same centre, the selected geometry increases mean annual production by 51.9 GWh and reduces wake loss by 0.89 percentage points. Under the supplied cost model, total capital expenditure is EUR 4,478.4 million and gross-yield levelized cost of energy is EUR 88.69/MWh. A separate planning case that allows for electrical losses, availability, and other delivery losses gives 4,849.5 GWh of net annual energy and a break-even delivered-energy price of EUR 97.62/MWh. The complete workflow is available at `https://github.com/MichaelIbrahim-GaTech/Sea-Wind-Phase-2-Code`; `train.py` fits the artifact and `inference.py` generates the forecast and siting outputs.

1 Task, data, and evidence boundary

1.1 Required outputs

The forecast output contains lower, median, and upper estimates for wind speed and direction at one, seven, and fourteen days. The scored footprint has 43,715 points. With eight inference windows, three lead times, and four forecast hours, the required row count is

$$43{,}715 \times 8 \times 3 \times 4 = 4{,}196{,}640.$$

The siting output specifies one centre and 55 turbine coordinates. Each turbine must remain inside the 15 km by 15 km layout box, every pair must be at least 1,420 m apart, and the centre depth must not exceed 50 m. The turbine type and total capacity are fixed by the rules [1, 2].

1.2 Declared data use

Training uses the official 2016–2020 high-resolution target, low-resolution reanalysis, supplied European Centre for Medium-Range Weather Forecasts High Resolution Forecast (HRES) tables, bathymetry, and static grids [3]. The definitive eight-window release is used only as an unlabeled inference input [4]. One external source is used for forecasting: causal GraphCast trajectories from the WeatherBench 2 archive. Here, causal means that each trajectory was initialized using information available at its issue time and contains no later observations. GraphCast is a global, ERA5-trained weather model and is sampled only from runs initialized at the relevant issue time [9, 10].

Historical target values are never available to `inference.py`. Reported evidence is separated into four categories:

- **Historical validation**: comparisons on reserved official target rows from 2016–2020.

- **Definitive forecast evaluation**: the six scores returned by the competition platform.
- **Deterministic audits**: row coverage, quantile ordering, siting legality, and energy bounds.
- **Scenario analysis**: economics and bidding calculations whose assumptions are stated explicitly.

1.3 Forecast score

Throughout the report, $\widehat{Q}_p$ denotes a conditional quantile at probability p. The speed interval is $[\widehat{Q}_{0.05}, \widehat{Q}_{0.95}]$, with median $\widehat{Q}_{0.50}$. For observation y, endpoints l and u, and nominal error probability $\alpha = 0.10$, the linear interval score is

$$S_\alpha(l, u; y) = (u - l) + \frac{2}{\alpha}(l - y)\mathbf{1}\{y < l\} + \frac{2}{\alpha}(y - u)\mathbf{1}\{y > u\}.$$

This proper score rewards narrow intervals only when they retain coverage [5, 6]. Direction uses the competition's circular counterpart because 0° and 360° represent the same direction. Lower scores are better. The primary score is rank based and is not the mean of the six raw scores.

2 Reproducible code and end-to-end execution

2.1 Code availability

The public repository at `https://github.com/MichaelIbrahim-GaTech/Sea-Wind-Phase-2-Code` contains `train.py`, `inference.py`, the pinned Python requirements, data-placement instructions, third-party licensing disclosures, this methodology report, and the final output archive. Competition datasets and external forecast fields are not redistributed. The README specifies where those inputs must be placed and gives the complete commands needed to regenerate the artifact and outputs.

The separation between training and inference is deliberate. Training may read labeled official rows from 2016–2020 and writes fitted parameters, validation decisions, the selected farm geometry, and the power-conversion policy. Inference reads only that artifact and unlabeled forecast inputs. It does not read historical targets, validation tables, or a previous `predictions.csv`. This makes the causal boundary inspectable in code rather than relying on a narrative claim.

2.2 Execution stages

Figure 1 gives the full computational path. The upper branch constructs forecast features, fits the probabilistic models, and retains optional corrections only after the historical checks described in Section 4.1. The middle branch screens legal centres before exact farm and layout replay. Both branches write one artifact. Inference reads that artifact and the unlabeled official windows, checks physical and structural constraints, and writes the two required outputs.

The production artifact contains 29 fitted estimators: nine lead-by-quantile speed models, six quantile refinements restricted to historically supported conditions, five context endpoint models, four circular direction residual models, one shared spatial direction model, two conditional direction endpoint models, and two component downscalers. An estimator here is one fitted statistical mapping with one narrowly defined output; the count is a component count, not 29 separate end-to-end weather forecasts. The nine common speed models are the Cartesian product of three forecast leads and three probabilities: 5%, 50%, and 95%. The six refinements and five context models make smaller corrections to speed interval endpoints where historical errors were repeatable. A condition is called historically supported only when the relevant calendar, location, and physical regime occurs in reserved observations and the proposed correction passes the paired checks in Section 4.1. Unsupported rows retain the preceding forecast. For direction, the circular residual models correct angular centres, the shared spatial model removes local directional bias, and the conditional endpoint models adjust angular uncertainty. The final two models predict eastward and northward wind separately on the 1.3 km grid. All components are compact, single-threaded statistical models with distinct jobs.

The repository README records the exact commands and numerical settings. Reproduction requires one call to `train.py`, followed by one call to `inference.py`. Four controls determine the balance between cost, sharpness, and stability. First, sampling one issue date every six days reduces training volume while retaining every season and all forecast hours. Second, conformal calibration expands speed intervals until reserved historical rows reach the 90% coverage target. Third, blend limits prevent a context or external trajectory from moving the forecast centre too far from the supplied atmospheric forecast. Fourth, a maximum circular half-width prevents the fourteen-day direction interval from expanding into an almost uninformative full circle. The saved artifact fixes these choices before inference.

3 Forecasting method

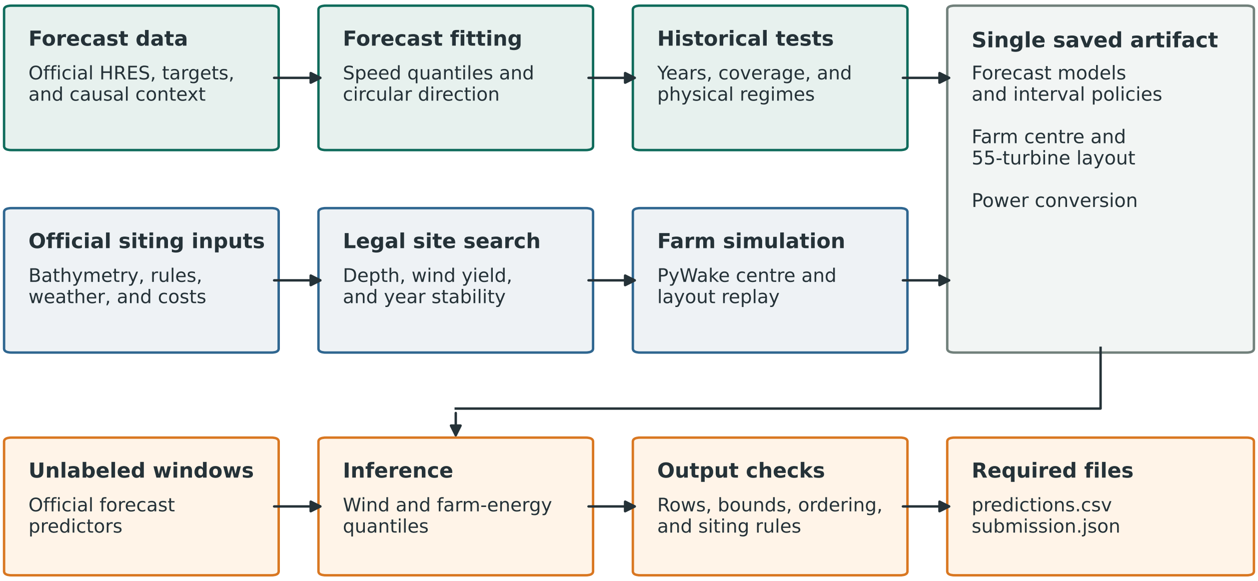

Figure 1: End-to-end generation pipeline. Labeled wind targets are confined to training and validation. Inference reads only the saved artifact and unlabeled official predictors.

3.1 Training examples and forecast anchors

One training example is one grid point at one issue date, target lead, and forecast hour. The four hours are 00, 06, 12, and 18 UTC. Supplied forecasts at one and seven days anchor the corresponding targets. For the fourteen-day target, the latest available supplied state at day ten is used as the atmospheric anchor. Training examples are sampled every six days to control memory and computation while retaining all seasons, locations, leads, and forecast hours.

Supplied speed s_f and meteorological direction θ_f are first converted to eastward and northward components,

$$u_f = -s_f \sin(\theta_f), \qquad v_f = -s_f \cos(\theta_f).$$

The minus signs follow the meteorological convention that direction identifies where the wind comes from. Each row contains u_f, v_f, s_f, latitude, longitude, forecast hour, and sine–cosine encodings of week of year. The speed model also receives s_f^2, $u_f v_f$, and sine–cosine encodings of forecast hour. These terms distinguish flow orientation, wind intensity, season, local position, and the daily cycle without introducing discontinuities at midnight or the end of the year.

3.2 Wind-speed quantiles

For each lead, separate LightGBM regressors estimate the 5th, 50th, and 95th conditional speed quantiles [7]. A model at probability τ minimizes the pinball loss

$$L_\tau(y, q) = \big(\tau - \mathbf{1}\{y < q\}\big)(y - q),$$

where y is observed speed and q is the predicted quantile. Directly fitting the three quantiles permits asymmetric and state-dependent uncertainty, which is important because forecast error generally increases with lead and wind regime. The nine lead-by-quantile models form the common speed forecast.

Interval width is calibrated on reserved historical rows. If l_i and u_i are the preliminary endpoints for validation observation y_i, the nonconformity score is

$$r_i = \max(l_i - y_i,\ y_i - u_i).$$

The finite-sample 90th-percentile value of r_i is added symmetrically to the interval endpoints. This conformal adjustment targets 90% marginal coverage without changing the median. Additional endpoint models and lookup corrections are considered only after this common calibration. They are retained when the paired interval score improves in every available validation year and no populated physical diagnostic group becomes worse.

The retained refinements have different roles. At one day, a lagged-context model adjusts speed endpoints in supported calendar and hour combinations. At seven days, a compact endpoint model and a small set of cell-level corrections use persistent forecast error that repeats across held years. At fourteen days, cell-specific endpoint transformations correct the systematic error of the day-ten atmospheric anchor. These branches alter interval endpoints rather than training a large atmospheric ensemble.

3.3 Circular wind direction

Direction is modeled through an angular residual rather than as an ordinary real-valued target. Let θ_f be the anchor direction and θ the observed direction. Their signed circular difference is

$$\delta = \mathrm{wrap}_{[-180,180)}(\theta - \theta_f).$$

Two regressors estimate $\sin\delta$ and $\cos\delta$. The predicted correction is reconstructed with

$$\widehat{\delta} = \mathrm{atan2}(\widehat{\sin\delta}, \widehat{\cos\delta}), \qquad \widehat{\theta} = \mathrm{wrap}_{[0,360)}(\theta_f + \widehat{\delta}).$$

This representation treats 359°and 1°as neighboring directions and avoids the artificial 358°error produced by ordinary subtraction [8]. Direction intervals are then represented by a centre and two circular endpoint displacements, so asymmetric uncertainty can be retained across north.

The one-day branch uses the supplied forecast as its main centre, then blends a lagged issue-time context correction and a causal GraphCast 1000 hPa trajectory where both historical folds support the change. The seven-day branch keeps the supplied seven-day field as its anchor. Five calendar-hour rules blend it with short-lead or weekly-vector information, one shared spatial residual model captures repeatable local bias, and two conditional endpoint models refine its circular interval. The fourteen-day branch combines weekly directional climatology with the latest supplied forecast signal. Only four cross-year-stable calendar-hour blends and seven seasonal centre rules are retained, and the final circular half-width is capped at 145°. Thus, longer lead time results in more conservative centre and width adjustments rather than a stronger extrapolation.

3.4 Terrain-aware downscaling

The probabilistic forecast is first produced on the coarse forecast grid. Two LightGBM regressors then map eastward and northward components to the 1.3 km target grid. Their inputs are bilinearly interpolated coarse u and v, interpolated speed, elevation, distance to shore, latitude, and longitude. Separate component models preserve wind-vector geometry and allow terrain and coastal position to affect the two components differently.

The downscaler is trained on official 2020 high-resolution targets at all four forecast hours, using every twentieth available day. At inference, each speed quantile is combined with the corresponding direction centre to form a coarse vector, the two components are downscaled, and speed and direction are reconstructed on the target grid. Sea-mask support is applied after prediction. Speed endpoints are sorted and clipped to satisfy $0 \le \widehat{Q}_{0.05} \le \widehat{Q}_{0.50} \le \widehat{Q}_{0.95}$, while every direction is wrapped to $[0, 360)$.

3.5 Final lead-specific workflow

Inference applies the fitted components in four stages. The common quantile models first produce a lower bound, median, and upper bound for speed at each lead. The retained lead-specific refinements then correct only the endpoints or centres for which historical support exists. Circular direction corrections are applied next, using vector operations throughout. Finally, the resulting wind vectors are downscaled to 1.3 km and converted back to ordered speed quantiles and wrapped directions. Short-lead branches use issue-time flow information directly, while the fourteen-day branch relies more heavily on seasonal and spatial regularity. The validation procedure below determines which optional corrections are permitted to enter this sequence.

3.6 Original contributions

The original contribution of this submission is the integration of four ideas into one constrained workflow. First, lead-specific probabilistic calibration is built around the supplied atmospheric forecasts instead of replacing them with an unrelated predictor. Second, angular centres and intervals are corrected in circular vector space, including conditional endpoint uncertainty that remains valid across 0°. Third, every optional correction is promoted by a conservative historical-support rule that requires improvement across the available years and diagnostic groups, rather than by aggregate score alone. Fourth, the same probabilistic wind field drives target-grid downscaling, farm siting, wake-aware layout design, and quantile bidding. This integration preserves a traceable path from meteorological inputs to the forecast, physical design, and financial decision.

4 Validation and forecast results

4.1 Validation protocol

Validation asks one practical question: does an added forecast component improve predictions on historical observations reserved for evaluation? The same four-step procedure is used for each speed and direction target:

1. Reserve one or more historical periods as validation data for the component being tested.
2. Produce two forecasts for exactly the same validation rows: one from the current pipeline without the new component and one from the candidate pipeline with it.
3. Score both forecasts with the competition interval score. A lower score indicates a better forecast. For speed, interval coverage must also remain at or above the stated 90% target.
4. Retain the component only if the total score improves and no available validation year or populated diagnostic group becomes worse. The diagnostic groups divide the rows by location, wind speed, direction, interval width, and signed error so that an average gain cannot conceal a clear localized loss.

The final inference labels are never read during this process. Their unlabeled predictors are used only to check whether a proposed correction would activate within conditions represented in the historical data. Table 1 reports one before-and-after component test for each scored forecast quantity. In every row, the baseline and candidate are evaluated on exactly the same historical observations; the only difference is the component named in the second column. Validation periods differ because the six components use different leads and activation conditions.

(a) Wind-speed component tests; lower interval score is better

Lead	Forecast change tested	Result on observations withheld from fitting
One day	Add a lagged-context correction to the lower and upper speed bounds	Two 174,860-row folds were tested. Both improved: 9.975 to 9.645 in 2019 and 13.040 to 11.468 in 2020. The coverage gate passed.
Seven days	Add a cross-year, grid-cell correction to the lower and upper speed bounds	Five complete years, 2016–2020, were tested. The score fell from 18.143 to 17.157; every year improved and coverage was 93.95%.
Fourteen days	Correct the speed interval constructed from the available ten-day forecast anchor	On 2,797,760 rows from 2019–2020, the score fell from 21.943 to 17.013. Both years improved and coverage was 91.75%.

(b) Wind-direction component tests; lower interval score is better

Lead	Forecast change tested	Result on observations withheld from fitting
One day	Blend a causal GraphCast trajectory into the predicted direction centre	Two chronological stress folds, 2018 and 2020, were tested. Both improved, by 2.329 and 7.385; the mean paired reduction was 4.857.
Seven days	Correct the lower and upper endpoints of the circular direction interval	Four complete held-out years, 2017–2020, were tested. Every year improved; reductions ranged from 2.007 to 2.785, with mean 2.331.
Fourteen days	Adjust circular interval width on February 25, where the correction is supported	On 349,720 rows covering all four hours in 2019 and 2020, every year-hour cell improved. Reductions ranged from 168.594 to 208.123.

Table 1: Historical before-and-after tests for components retained in the final forecast. Each candidate is compared with the preceding pipeline on exactly the same withheld observations, while changing only the component named in the middle column. Speed results report absolute scores before and after the change. Direction results report paired reductions because the corrections can apply only to supported subsets. The fourteen-day direction result is local to four February 25 forecast hours; results from different rows therefore must not be added together.

The speed rows show that lower interval scores were obtained without reducing coverage below the nominal 90% target. The direction rows emphasize consistency across years because their accepted corrections act only on historically supported subsets. The large fourteen-day direction reduction is a local result for four active forecast hours, not a reduction of the complete fourteen-day score.

Two examples clarify the selection rule. A stronger one-day external speed blend improved its mean score but failed localized issue-time checks and was rejected. A separate fine-grid speed-residual model failed its transfer gate and is not present in the production artifact. These rejected branches matter because they show that aggregate development score alone did not determine the final pipeline.

4.2 Interpretation of the fitted signal

LightGBM records how much each tree split reduces the training loss. Summing this reduction by input family and converting the totals to percentages gives normalized split gain, which is used here as a descriptive audit rather than a causal explanation or a sensitivity test. At one day, the supplied forecast-speed variables account for 50.6% of normalized gain; location, vector components, and seasonal terms contribute 14.3%, 11.5%, and 11.3%. At fourteen days, direct forecast speed contributes only 5.4%, while seasonal terms, location, and wind-vector components together contribute 76.5%. This shift is consistent with the information boundary: direct forecast guidance is strongest at short lead, whereas the long-lead model relies more heavily on climatological and spatial structure.

Activation fractions provide a second, model-independent interpretation. The accepted one-day context correction modifies 61.87% and 74.97% of rows in the two retained validation folds; the seven-day correction is deliberately sparse at 6.25%; and the fourteen-day endpoint policy acts on 84.38% of its evaluated rows. These fractions explain why a small seven-day aggregate gain can still be credible and why the fourteen-day policy required stronger year and diagnostic-group checks.

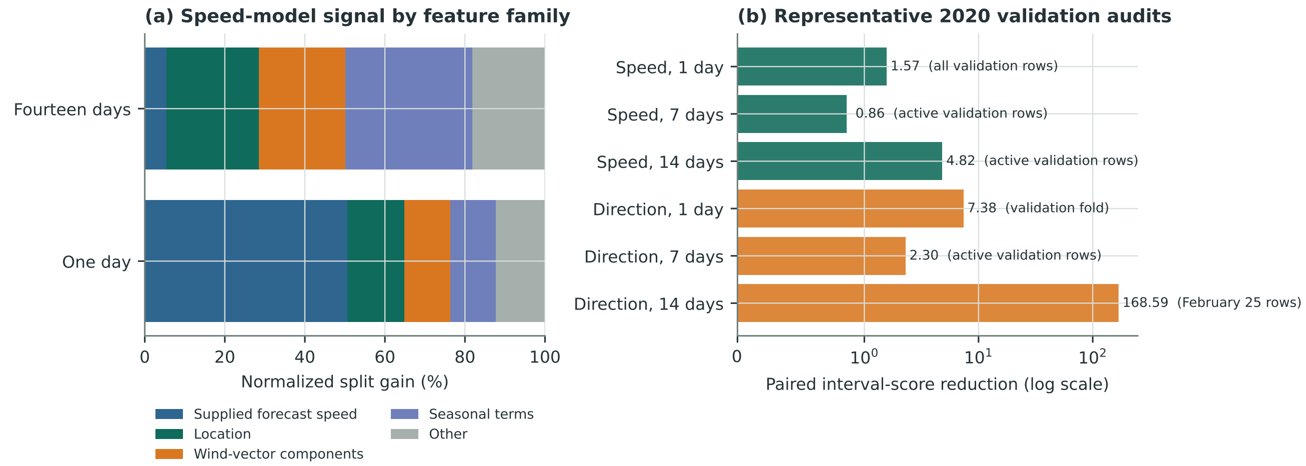

Figure 2: Model interpretation and validation audit. Panel (a) groups normalized split gain by feature family; it describes model use rather than causality. Panel (b) shows a representative paired reduction on 2020 validation rows for each target branch and its component-specific activation mask. The logarithmic axis makes the sparse February-only fourteen-day direction audit visible without hiding the other five; magnitudes are not compared across masks.

4.3 Definitive evaluation

The competition scorer matched 4,196,640 of 4,196,640 rows. Table 2 reports the complete definitive result.

Interval score	One day	Seven days	Fourteen days
Wind speed	9.511	17.124	16.835
Wind direction	76.400	335.571	315.183
Platform primary score			1.497276

Table 2: Definitive forecast scores returned by the competition platform. Lower values are better; the primary score is rank based.

The result is uneven. One-day direction is the strongest directional component, whereas seven- and fourteen-day direction remain the largest forecast weakness. The report does not infer calibration or generalization from the platform score alone.

5 Wind-farm siting and power conversion

5.1 Centre selection

The centre search first evaluates gross wind on the supplied grid because wake simulation at every cell would be unnecessarily expensive. Of 37,835 weather cells screened, 23,406 pass the geographic and depth rules. Fourteen candidates representing high mean wind, stable worst-year wind, balanced performance, and depth margin receive exact five-year PyWake replay. The final run retains the selected reference centre because no challenger improves mean and worst-year capacity factor while also satisfying the neighborhood, wake, depth, and economic gates. This is a robust selection rule, not proof of a global optimum.

5.2 Layout selection

At the selected centre, 16 valid layouts are replayed with the prescribed IEA 22 MW turbine and Bastankhah-Gaussian wake model [11]. The final geometry comes from constrained coordinate refinement and uses the available square while preserving spacing. Figure 4 shows the 55 submitted positions.

The legal audit is deterministic and is summarized in Table 3. The centre is inside the allowed zone and at 44.14 m depth. The layout uses almost all of the permitted north–south span, so coordinates are retained at full numerical precision in `submission.json`; rounding the outer coordinates would unnecessarily consume the remaining 0.06 m margin.

The selected centre is 69.07 km from the nearest coast in the supplied distance layer. Distance and depth enter the cost model through export-cable and foundation premiums. Economics is therefore used as a secondary discriminator between near-equal capacity-factor candidates; it does not displace a materially better yield candidate.

Table 4 makes that trade-off explicit. The balanced alternative is shallower and closer to shore, but the selected design gains 0.67 percentage points in mean capacity factor and 0.71 percentage points in the worst year. The

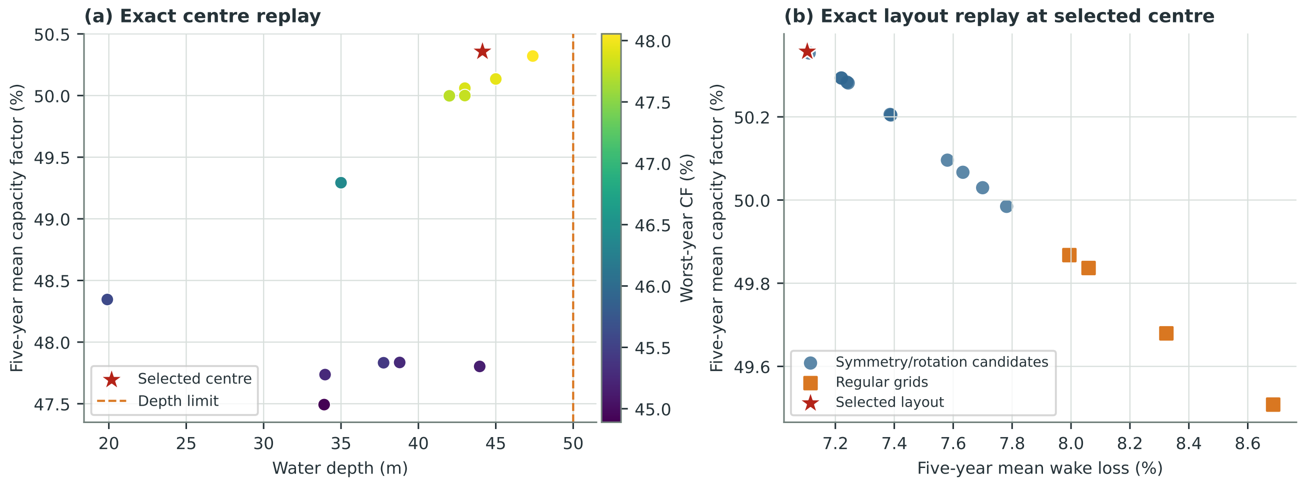

Figure 3: Exact five-year siting evidence. Panel (a) shows the 14 shortlisted legal centres; colour records worst-year capacity factor, so the selected point combines high mean yield with cross-year resilience while retaining depth margin. Panel (b) shows all 16 valid layout candidates at the selected centre; moving upward and left means more yield and less wake loss.

Constraint	Observed	Requirement	Margin
Turbine count	55	exactly 55	0
Installed capacity	1,210 MW	exactly 1,210 MW	0
Centre depth	44.14 m	at most 50 m	5.86 m
Maximum $\lvert x \rvert$	7,495.61 m	at most 7,500 m	4.39 m
Maximum $\lvert y \rvert$	7,499.94 m	at most 7,500 m	0.06 m
Minimum pairwise spacing	1,554.64 m	at least 1,420 m	134.64 m

Table 3: Deterministic legality audit of the submitted centre and coordinates.

lowest-LCOE alternative reduces modeled capital cost by EUR 773.5 million, but gives up 2.43 percentage points of mean capacity factor and 3.01 percentage points in the worst year. Because official siting rank is based on capacity factor, the lower-cost candidates are retained as financial comparators rather than selected designs.

Audited design	Depth (m)	Coast (km)	Mean CF	Worst CF	CAPEX (EUR m)	LCOE (EUR/MWh)
Selected	44.1	69.1	50.36%	48.18%	4,478.4	88.7
Balanced alternative	41.0	57.1	49.68%	47.47%	4,288.5	86.2
Lowest-LCOE alternative	30.9	21.3	47.93%	45.18%	3,704.8	78.5

Table 4: Yield and economic trade-off among the selected design and two audited legal alternatives. All values use five-year PyWake replay and the supplied Phase 2 cost model.

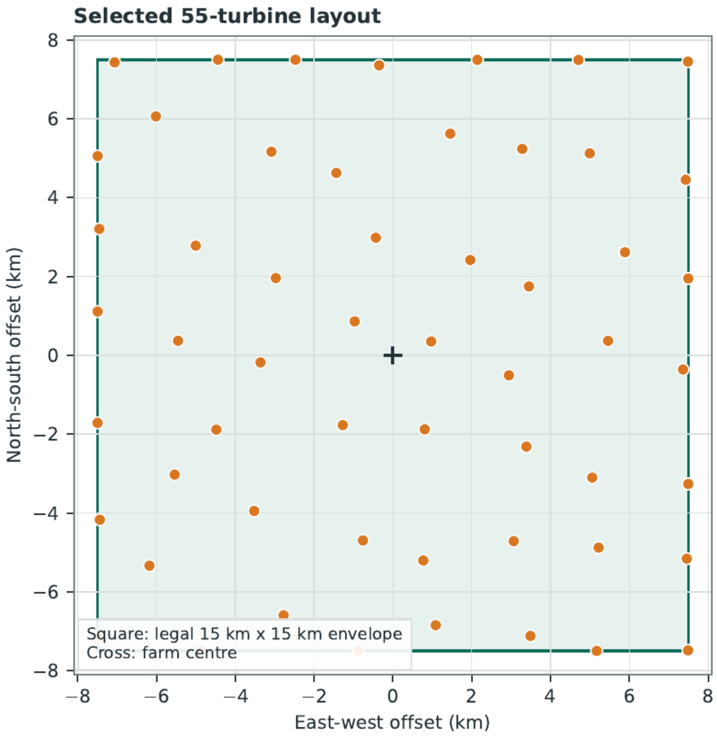

Figure 4: Submitted turbine coordinates relative to the farm centre. The dashed square is the permitted layout envelope.

Coordinate refinement evaluated 340 legal moves and accepted 26. Against the pre-refinement geometry, the final coordinates improved mean capacity factor by 0.013 percentage points, worst-year capacity factor by 0.019 percentage points, and every individual training year. In a 2,000-replicate monthly block bootstrap, every replicate retained a positive capacity-factor change; the first-percentile change was 0.009 percentage points. The layout also passed the mean-yield, worst-year-yield, and wake gates at all 31 nearby legal centres tested. These small but consistently signed changes support robustness; they are not presented as a material project-level yield increase.

Table 5 isolates the value of the layout by holding the centre and five-year weather replay fixed. Relative to a valid rectangular 7-by-8 arrangement with one unused position, the submitted geometry adds 51.9 GWh/year on average and reduces wake loss by 0.89 percentage points.

Layout at selected centre	Mean CF	Worst CF	Mean AEP (GWh/year)	Mean wake
Valid rectangular 7-by-8	49.87%	47.73%	5,285.8	7.99%
Submitted geometry	50.36%	48.18%	5,337.7	7.10%
Change	+0.49 pp	+0.45 pp	+51.9	−0.89 pp

Table 5: Same-centre, same-weather comparison that isolates the submitted layout. CF is capacity factor; pp denotes percentage points.

Replay year	AEP (GWh)	Capacity factor	Wake loss
2016	5,106.9	48.18%	7.31%
2017	5,824.1	54.95%	6.93%
2018	5,279.1	49.80%	7.00%
2019	5,327.4	50.26%	7.31%
2020	5,150.7	48.59%	6.97%
Mean	5,337.7	50.36%	7.10%

Table 6: Exact annual PyWake replay of the selected site and layout using the supplied historical wind.

The annual spread is material: capacity factor ranges from 48.18% to 54.95%, while wake loss remains between 6.93% and 7.31%. A directional stress test shifts the complete wind rose from $-45°$ to $+45°$; capacity factor remains between 50.23% and 50.38%, while maximum wake loss is 7.32%. This test shows that the irregular geometry is not narrowly tuned to one wind-rose orientation.

A second benchmark evaluates the combined centre-and-layout decision on public training data. A plain 7D grid at the public reference centre, snapped to 53.4982°N and 1.4928°E, gives 47.08% mean capacity factor, 4,989.9 GWh mean AEP, and 9.70% mean wake loss under the same five-year replay. The selected design gives 50.36%, 5,337.7 GWh, and 7.10%, respectively. This is explicitly a public-data proxy, not the organizer's hidden real-wind baseline; it is included because its inputs and simulation path are reproducible.

5.3 Farm-energy quantiles

For each inference time, `inference.py` extracts wind quantiles at the selected centre, applies the prescribed power-law shear from 125 m to the 170 m hub height, and runs the submitted geometry through the turbine power curve and wake model. The three paths are sorted after simulation and clipped to the physical six-hour range $[0, 7{,}260]$ MWh. The final JSON contains 96 values for each quantile. Their means are 738.9, 3,001.8, and 6,735.1 MWh for the lower, median, and upper paths.

This construction guarantees consistency with the submitted wind forecast and farm geometry. It does not guarantee calibrated farm-power coverage because a common median direction is used for all three speed paths and the joint speed–direction distribution is not modeled.

6 Economics and forecast value

6.1 Construction and operating cost

The supplied Phase 2 cost model is evaluated at the selected centre. Table 7 shows the complete decomposition used in the report. Cost intensity means component cost divided by the farm's full installed capacity of 1,210,000 kW. For example, 1,030 EUR/kW corresponds to EUR 1,246.3 million for the 1.21 GW farm. Depth and export distance are the site-dependent terms; no cost item is replaced with an external estimate.

Capital component	Cost intensity (EUR/kW)	Total (EUR million)
Turbines and towers	1,030.00	1,246.3
Foundations, including depth premium	812.10	982.6
Array electrical system	350.00	423.5
Export cable, including distance premium	552.56	668.6
Installation	350.00	423.5
Development	310.00	375.1
Contingency	296.47	358.7
Total capital expenditure	**3,701.13**	**4,478.4**

Table 7: Construction-cost result from the supplied Phase 2 cost model. Costs in the middle column are normalized by 1.21 GW of installed nameplate capacity.

Annual operating expenditure is EUR 123.1 million. Using mean historical energy, the supplied model gives a levelized cost of energy of EUR 88.69/MWh, comprising 65.63 EUR/MWh of capital recovery and 23.06 EUR/MWh of operation. Replacing mean energy with the lowest historical year raises the result to 92.70 EUR/MWh. This four-euro range is the direct financial consequence of the recorded interannual wind variability under the supplied assumptions. These figures use gross PyWake energy, as required for the competition siting comparison. Section 6.2 separately applies delivery losses before calculating revenue.

Two independent references provide a plausibility check, not a replacement cost model. The 2025 *Guide to an Offshore Wind Farm* uses a representative 1 GW UK project at 50 m depth and 75 km distance and reports capital cost of about GBP 3.47 million/MW and annual operating cost of GBP 85,000/MW [15]. The selected site's EUR 3.70 million/MW capital cost and EUR 101,721/MW-year operating cost are of comparable order. A direct numerical match is neither expected nor claimed because currencies, price years, turbine ratings, grid boundaries, and contracting scope differ. NREL's 2025 United States fixed-bottom assessment places most sites above USD 6,000/kW and operating cost above USD 83/kW-year; it also finds that distance from coast, port, and interconnection increases both costs [17, 16]. The higher United States capital estimate shows why geographic and accounting boundaries matter. Taken together, the literature supports the model's dependence on depth and distance but validates only order of magnitude, not project bankability.

6.2 Historical market context, net energy, and viability

The five-year gross AEP is 5.338 TWh. Public Energy-Charts data report 100.39 TWh of Dutch load in 2022, so the modeled farm output is equivalent to 5.32% of that annual quantity [20]. The organizers did not specify a grid landing point or bidding zone. Dutch data are therefore used only as a transparent adjacent North Sea reference with hourly observations covering the forecast timestamps, not as a claim that the farm would settle in the Netherlands. The 2022 Dutch day-ahead mean was 241.92 EUR/MWh and the median was 217.00 EUR/MWh. This exceptional year provides historical stress context rather than a long-term price forecast. As an independent offshore-contract reference, the United Kingdom's Allocation Round 7 cleared fixed-bottom projects at GBP 89.49–91.20/MWh in

2024 prices [23]. That contract benchmark has a different currency, support structure, and delivery boundary, so no direct equivalence is claimed.

For the 32 available one-day definitive blocks, the median farm-energy forecast totals 121.193 GWh. Aligning each six-hour block with the Dutch hourly series gives a gross forecast value of EUR 23.751 million and an energy-weighted capture price of 195.98 EUR/MWh, compared with an unweighted 229.93 EUR/MWh over those hours. This calculation demonstrates timestamp alignment and price weighting. It is not a realized revenue result because actual farm production and imbalance settlements are unavailable.

The PyWake result already includes wake loss but does not represent all energy delivered to the grid. For financial planning, the gross AEP is therefore reduced by three explicit factors. The NREL offshore resource methodology uses 96% availability, a 2% allowance for other performance, environmental, and curtailment losses, and a depth-and-distance relationship for electrical loss [22]. At 44.14 m depth and 69.07 km export distance, that relationship gives 3.43% electrical loss. Multiplying the remaining fractions gives a net-delivery factor of 90.85% and net AEP of 4,849.5 GWh/year. This adjustment is used only for the planning analysis below; the official siting results remain the gross PyWake values.

Table 8 makes the central case and every tested range explicit. The 25-year life follows the NREL reference design. NREL reports a 4.01% real weighted average cost of capital for offshore wind; 6% is used here as a more conservative constant-price planning rate and is tested from 4% to 8% [21]. The 70–110 EUR/MWh price range is a stress band, not a forecast. Its central value is near the supplied gross-yield LCOE and of similar numerical order to the separate UK contract benchmark, subject to the differences noted above.

Input	Central value	Tested range	Basis
Gross AEP	5,337.7 GWh/year	5,106.9–5,824.1	Five annual PyWake replays
Availability	96%	95–98%	NREL reference and O&M range
Electrical loss	3.43%	Fixed	NREL depth-distance relationship
Other delivery loss	2.00%	Fixed	NREL aggregate allowance
Net AEP	4,849.5 GWh/year	4,640–5,290	Gross annual AEP after delivery factors
Capital expenditure	EUR 4,478.4 million	±10%	Supplied model and sensitivity
Operating expenditure	EUR 123.1 million/year	±10%	Supplied model and sensitivity
Energy price	90 EUR/MWh	70–110	Constant real-price stress band
Real discount rate	6%	4–8%	Conservative case around NREL reference
Operating life	25 years	20–30	NREL reference and sensitivity

Table 8: Inputs for the net-delivery financial scenarios. A range means that the parameter is varied while other central assumptions remain fixed.

With capital expenditure C_0, net annual energy E_{net}, annual operating cost O, constant real energy price P, real discount rate r, and operating life T, the pre-tax calculation is

$$\text{NPV}(P) = -C_0 + \sum_{t=1}^{T} \frac{PE_{\text{net}} - O}{(1+r)^t}.$$

The break-even delivered-energy price is 97.62 EUR/MWh. Table 9 shows the three price cases. At 90 EUR/MWh the project has a 4.86% internal rate of return and does not recover its capital on a discounted basis within 25 years. Positive NPV is reached in the 110 EUR/MWh case, with discounted payback after 18.3 years.

Price (EUR/MWh)	Annual revenue (EUR million)	NPV (EUR million)	IRR	Discounted payback
70	339.5	−1,712.3	1.51%	Beyond 25 years
90	436.5	−472.4	4.86%	Beyond 25 years
110	533.4	767.5	7.74%	18.3 years

Table 9: Investment results using net annual energy, a 6% real discount rate, and a 25-year operating life.

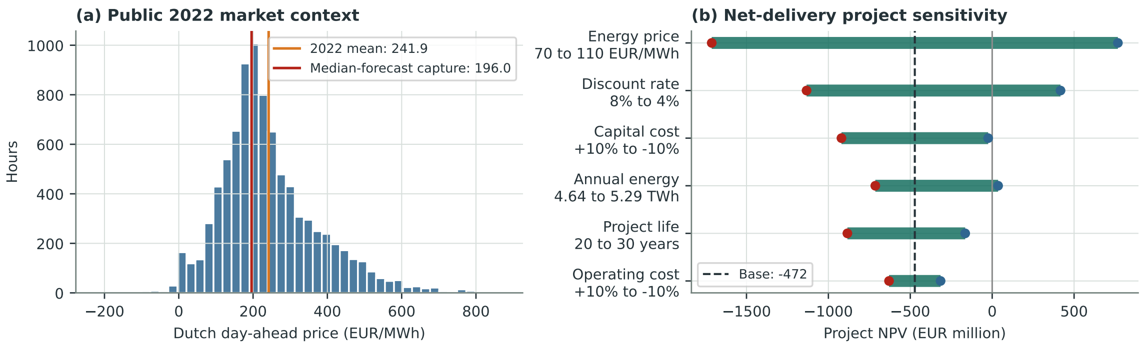

Figure 5: Historical market context and net-delivery project sensitivity. Panel (a) shows hourly Dutch day-ahead prices in 2022 and the forecast-weighted capture price for the 32 one-day blocks. Panel (b) varies one assumption at a time around the 90 EUR/MWh, 6%, 25-year case. Every tested range is printed beside its driver; endpoints are scenario bounds, not confidence intervals.

The supplied operating-cost value is an aggregate allowance rather than a component-level maintenance model. Offshore operations combine scheduled work, unscheduled repair, marine access, insurance, and asset management; access restrictions can place availability in the 95–98% range [18]. NREL identifies condition-based maintenance, remote inspection, automated repair, improved personnel transfer, and better weather forecasting as credible improvement mechanisms [19]. The practical strategy is to monitor vibration, temperature, oil condition, alarms, and power-curve residuals; update component failure hazards; and schedule vessel work in forecast low-wind access windows. At 90 EUR/MWh, changing availability from 95% to 98% increases net AEP from 4,799.0 to 4,950.6 GWh/year and improves NPV from EUR -530.5 to -356.2 million. This range quantifies the exposure that maintenance must manage without inventing a component-level saving that the available condition and repair data cannot support.

6.3 Quantile bidding

If shortfall and excess penalties are π^+ and π^-, respectively, the cost-minimizing bid is the predictive quantile [12]

$$\tau^* = \frac{\pi^-}{\pi^+ + \pi^-}.$$

For the supplied illustrative penalties of 90 EUR/MWh for shortfall and 30 EUR/MWh for excess, $\tau^* = 0.25$. The 25th-percentile energy bid is obtained by linear interpolation between the literal `predicted_q05` and `predicted_q50` arrays. Across the 32 one-day six-hour blocks, its mean is 2,487.1 MWh and the total commitment is 79.59 GWh.

To quantify the information value without pretending that outcomes are observed, the submitted 5th, 50th, and 95th percentiles define a piecewise-linear predictive quantile function; the two five-percent tails are linear continuations clipped to the physical range. Integrating the asymmetric loss over that distribution gives an expected imbalance cost of EUR 2.504 million for a median bid and EUR 1.880 million for the 25th-percentile bid. The quantile policy therefore reduces ex-ante expected imbalance cost by EUR 0.624 million, or 24.9%, across the available blocks under the stated penalties. This is a reproducible model-based value calculation, not a realized profit backtest. Realized farm production and imbalance settlement prices for the definitive blocks are unavailable.

The conclusion is not specific to one penalty ratio. Table 10 repeats the same calculation for four asymmetric policies. As the shortfall penalty rises relative to the excess penalty, the optimal bid moves to a lower predictive quantile. In every tested case, this policy has lower expected cost than bidding the median.

Shortfall (EUR/MWh)	Excess (EUR/MWh)	Optimal quantile	Expected cost (EUR million)	Reduction from median bid
60	30	0.333	1.672	11.1%
90	30	0.250	1.880	24.9%
120	30	0.200	2.004	35.9%
90	45	0.333	2.508	11.1%

Table 10: Ex-ante bidding sensitivity across the 32 one-day six-hour blocks. Expected costs use the submitted predictive intervals and the same piecewise-linear quantile approximation in every row.

7 Frugality, transparency, and verification scope

7.1 Compute footprint

Training and siting replay took 4.840 CPU-hours; inference and archive generation took 0.272 CPU-hours. No GPU was used, and every estimator and numerical library was restricted to one thread. Table 11 converts wall-clock time using the Intel Core i7-7700HQ processor's nominal 45 W thermal design power [13]. This CPU-package proxy uses a declared carbon sensitivity of 0.45 $kgCO_2e/kWh$ [14].

Stage	CPU-hours	GPU-hours	kWh	$kgCO_2e$
Training, validation, and siting	4.840	0	0.218	0.098
Inference and packaging	0.272	0	0.012	0.006
Total	**5.112**	**0**	**0.230**	**0.104**

Table 11: Measured runtime and bounded CPU-package energy and carbon estimates.

The estimate excludes memory, storage, display, networking, cooling, and production of the public GraphCast archive.

7.2 External resource disclosure

GraphCast is the only external forecast resource. Its 37-level model is documented as trained on ERA5 from 1979 through 2017; the implementation is Apache-2.0 and the weights are CC BY-NC-SA 4.0 [10]. Only causal forecast fields are sampled, and neither fields nor weights are redistributed. No other third-party forecast is used. Energy-Charts data, licensed CC BY 4.0, are used only for the report's retrospective market context [20].

7.3 LLM and agent methodology

An OpenAI Codex coding agent powered by GPT-5 was used as an engineering assistant for drafting and refactoring Python code, automating experiment execution, and improving technical documentation. It was not part of the forecasting system, supplied no meteorological data or forecasts, and was not used during inference.

All generated code and proposed changes were reviewed before execution. Experimental objectives, allowable inputs, model selection, siting and economic assumptions, and final artifacts remained under human control. Suggestions that did not comply with the rules or failed local cross-validation were rejected. The agent's principal contribution was efficient experiment auditing and consolidation of the accepted method into the reproducible two-file pipeline.

7.4 Verification scope

Exact output checks cover row count, quantile order, turbine count, spacing, footprint, and centre depth. Historical validation and robustness tests assess forecast and siting transfer, while stated scenarios bound project value.

Forecast transfer. Chronological, worst-year, and diagnostic-group tests reduce dependence on one favorable period or narrow subset. Additional causal atmospheric information is the clearest extension for long-range direction.

Siting transfer. Five annual replays, nearby-centre tests, bootstrap comparisons, and wind-direction rotations test sensitivity to year and wind-rose orientation. Development would add site surveys and on-site wind observations.

Farm-energy quantiles. Estimates are ordered, bounded, and generated from the submitted forecast and layout. Joint speed-direction modeling would improve power tails when matched observations become available.

Economic scope. The 2% other-loss factor is aggregate, not a site-specific curtailment model. Construction schedule, financing, taxation, maintenance records, grid terms, and residual value require project-specific evidence, so the financial results remain planning scenarios.

8 Conclusion

The solution integrates probabilistic wind prediction, legal farm design, and farm-energy quantiles in a reproducible two-file pipeline. Evidence includes full definitive forecast coverage, paired validation for all six targets, a legal 55-turbine layout with lower wake loss than the same-centre grid, and cross-year siting tests. The financial analysis identifies a EUR 97.62/MWh break-even delivered-energy price and separates the effects of market price, financing, energy, cost, availability, and life. Quantile bidding then converts forecast uncertainty into an operational decision under several penalty ratios. Code and generation instructions are available in the public repository.

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